CLASS 8 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 3

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Two persons pull a rope with 100 N each in opposite directions. The net force on the rope is: 1
(a) 50 N **(b)** 0 N **(c)** 100 N **(d)** 200 N

2. A brick of weight 20 N has faces of areas 0.1 square metre and 0.04 square metre. The minimum pressure it can exert when resting on a face is: 1
(a) 200 Pa **(b)** 500 Pa **(c)** 0.2 Pa **(d)** 2 Pa

3. Which of the following sets are in the right order of increasing space between particles? 1
(a) solid, liquid, gas **(b)** gas, liquid, solid
(c) liquid, solid, gas **(d)** gas, solid, liquid

4. A student heated a mixture of iron and sulphur strongly and got a black solid that is not magnetic. The black solid is a: 1
(a) element **(b)** compound **(c)** alloy **(d)** mixture

5. How much water is needed to make 500 g of a 5 per cent salt solution? 1
(a) 25 g **(b)** 475 g **(c)** 500 g **(d)** 450 g

6. An electroscope's leaves collapse when a charged rod is touched by a finger because: 1
(a) charge flows to earth **(b)** leaves become heavy
(c) a magnet acts **(d)** air enters

7. The sea breeze occurs because: 1
(a) sea is warmer at noon **(b)** land is cooler than sea in the daytime
(c) land is warmer than sea in the daytime **(d)** sea is frozen

8. Water at 4 degrees Celsius and 90 degrees Celsius: the faster-moving particles are in: 1
(a) water at 4 degrees Celsius **(b)** cannot be said
(c) equal in both **(d)** water at 90 degrees Celsius

9. Which element is both lustrous and brittle? 1
(a) iron **(b)** silicon **(c)** copper **(d)** gold

10. Which of these settles on standing? 1
(a) salt in water **(b)** sand in water **(c)** milk **(d)** sugar in water

11. A magnet is cut into two pieces. Each piece will: 1

(a) lose magnetism **(b)** be a north pole only
(c) be a south pole only **(d)** have two poles

12. The speed of a cyclone's wind is highest in the: 1

(a) the centre **(b)** eye **(c)** outer edge only **(d)** eye wall

13. Why does a closed container of water not appear to lose water? 1

(a) pressure is low
(b) vapour cannot escape; condensation balances evaporation
(c) no evaporation occurs
(d) water is solid

14. The cooking utensils are made of metals because they are: 1

(a) poor conductors **(b)** lighter than water
(c) brittle **(d)** good conductors of heat

15. The solubility of salt at 20 degrees Celsius is 36 g per 100 g of water. If 40 g is added to 100 g of water at 20 degrees Celsius, then: 1

(a) 4 g remains undissolved **(b)** 36 g remains
(c) 40 g remains **(d)** all dissolves

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why does rubbing hands together warm them? 2

17. Why do winds blow from high to low pressure? 2

18. Differentiate between vapour and gas. 2

19. Why is the formula of water not written as HO? 2

20. Why does a lemon-water solution not scatter light but milk does? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Describe three methods for separating components of solutions. 3

22. Explain how a force of 20 N and a force of 12 N can produce both a net force of 32 N and one of 8 N on the same body. 3

23. The pressure at the bottom of a pond is more than at the top. Explain with an activity. 3

24. Explain why the rate of diffusion in gases is more than in liquids. 3

OR

How can you distinguish a mixture of iron and sulphur from iron sulphide? Give three tests. 3

25. How much sugar must be dissolved in 360 g of water to prepare a 10 per cent solution? 3

OR

How do you show that friction depends on the load? Describe a method. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Discuss the use of metals and non-metals in the human body, daily life and industry with five examples. 5

OR

Explain the changes that occur when a saturated solution of KNO3 at 60 degrees Celsius is cooled to 20 degrees Celsius. 5

27. Explain, with reasons, which type of friction acts in: 5
(a) a car just starting
(b) a box sliding
(c) a rolling ball
(d) a parked scooter on a slope.

OR

Explain pressure in liquids using the example of a dam and water jets. Mention the dependence on depth and direction. 5

END OF QUESTION PAPER — **Total: 50 Marks**

MASTERMIND